Delete Node in a BST (medium)

Problem Statement

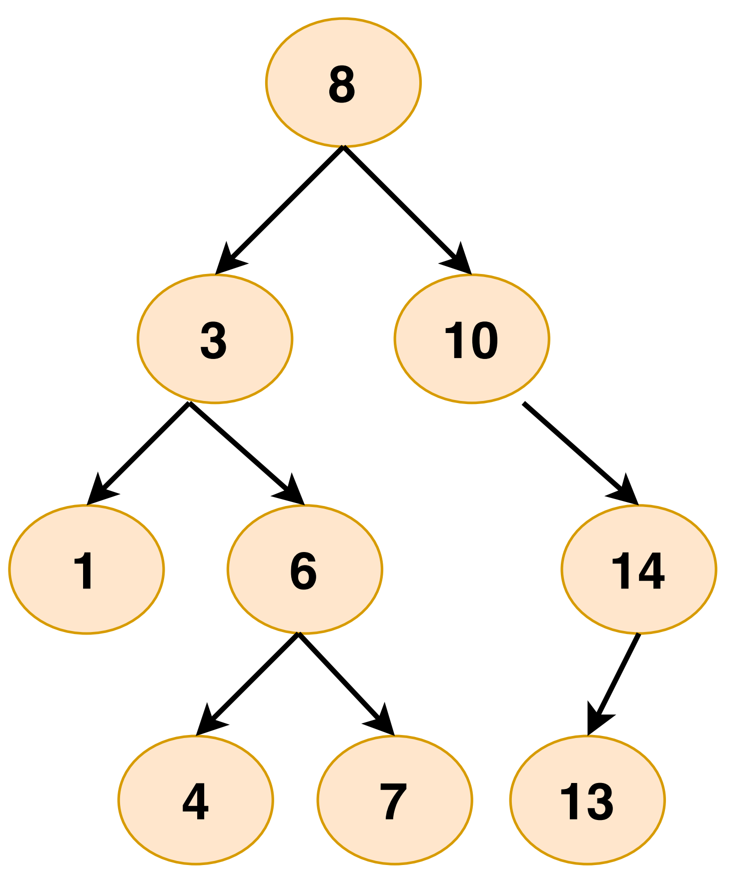
Given a `root` node of the binary search tree, and an integer value `key`, return the `root` of the updated BST after `deleting` the node having a value equal to `key`.

If BST doesn't contain the node having value equal to `key`, return the same binary tree.

Examples

Example 1:

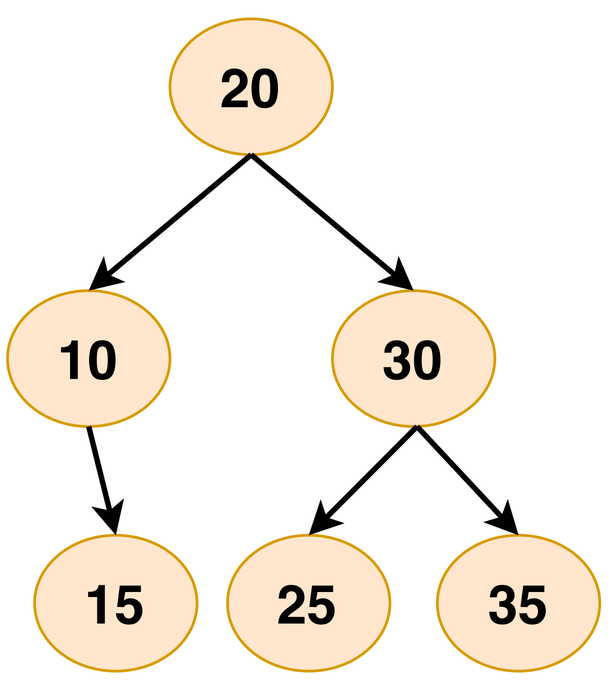
- **Input**: `[8, 3, 10, 1, 6, null, 14, null, null, 4, 7, 13]`, **key** = `3`



- **Expected Output**: `[8, 4, 10, 1, 6, null, 14, null, null, null, 7, 13]`
- **Justification**: After removing the node with value `3`, its right subtree's smallest value (`4`) replaces it to maintain the BST structure.

Example 2:

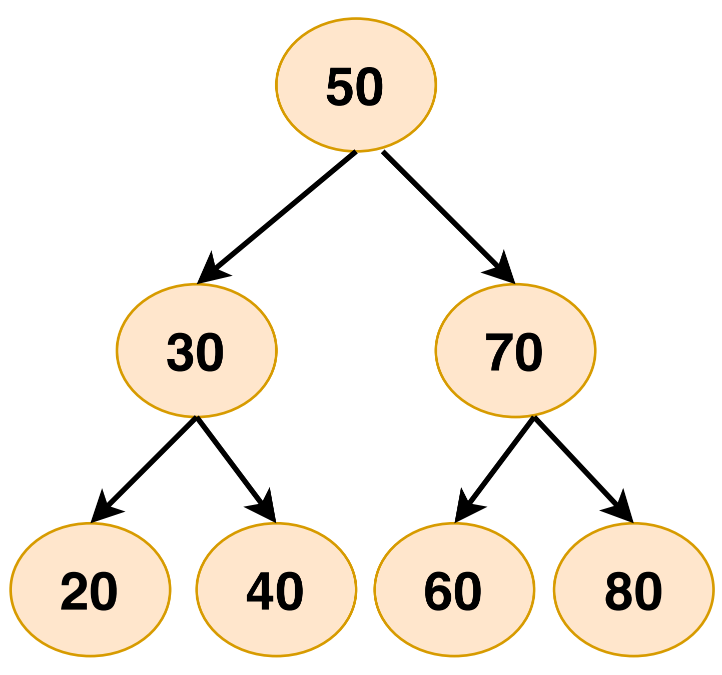
- **Input**: `[20, 10, 30, null, 15, 25, 35]`, **key** = `10`



- **Expected Output**: `[20, 15, 30, null, null, 25, 35]`
- **Justification**: The node `10` is removed and replaced by its right child `15` to keep the tree's integrity.

Example 3:

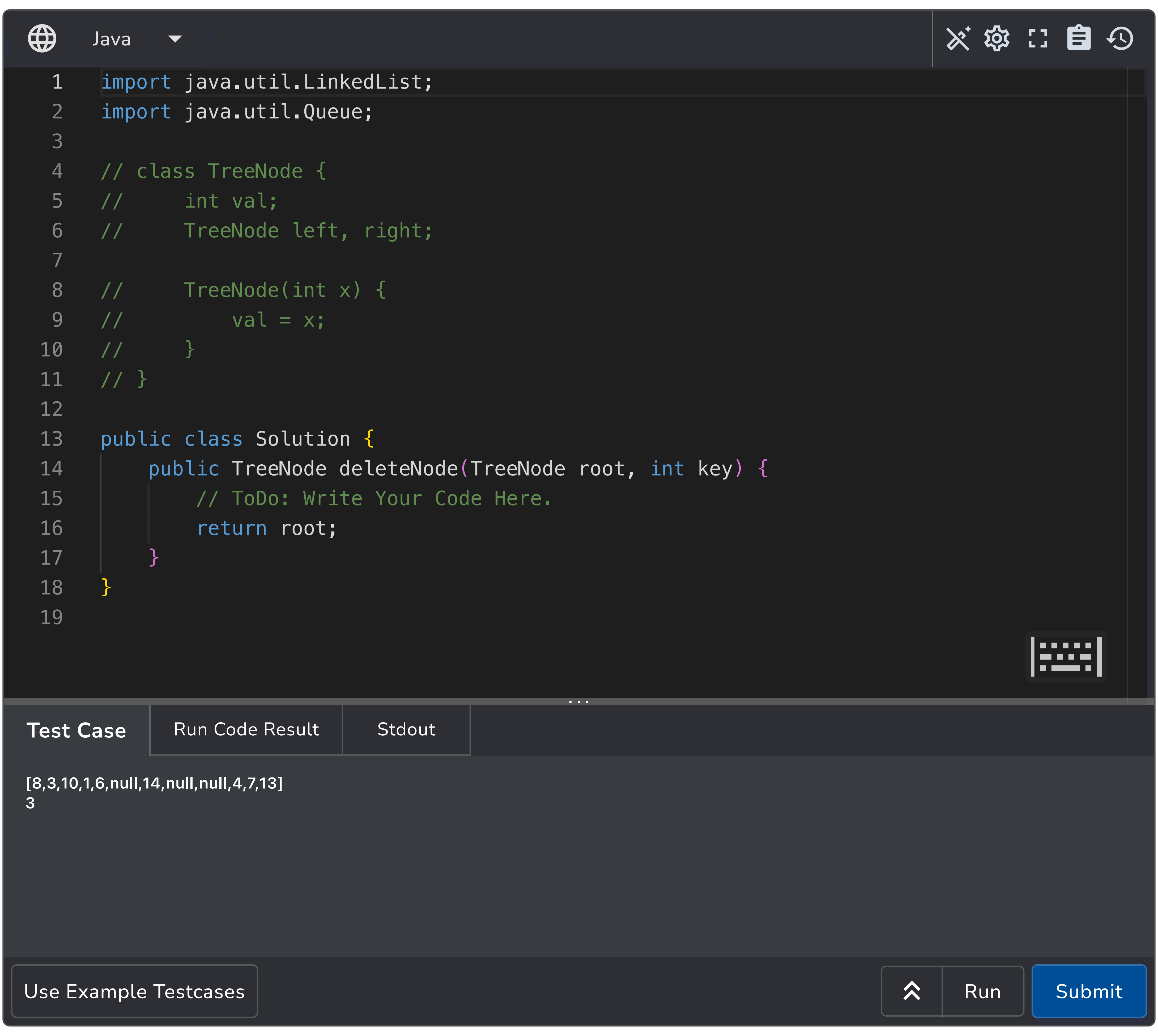
- **Input**: `[50, 30, 70, 20, 40, 60, 80]`, **key** = `10`



- **Expected Output**: `[50, 30, 70, 20, 40, 60, 80]`
- **Justification**: The BST doesn't contain the node with value 10. So, it returns the original BST.

Try it yourself

Try solving this question here:



```java
import java.util.LinkedList;
import java.util.Queue;

// class TreeNode {
//     int val;
//     TreeNode left, right;

//     TreeNode(int x) {
//         val = x;
//     }
// }

public class Solution {
    public TreeNode deleteNode(TreeNode root, int key) {
        // ToDo: Write Your Code Here.
        return root;
    }
}
```

```
[8,3,10,1,6,null,14,null,null,4,7,13]
3
```

← Previous
Solution: Spiral Matrix

Next →
Solution: Delete Node In A BST

☐ Mark as Completed